Piyush Singh

Data Scientist

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Summary

Data Scientist with 2 years of experience developing LLM-powered and machine learning systems for NLP, computer vision, and analytics. Skilled in Python, PyTorch, LangChain, and cloud platforms (AWS, GCP, Docker). Passionate about building ethical and scalable AI solutions that bridge research with real-world applications.

Experience

AI Consultant — Feb. 2025 – May 2025
DeHaat — *Remote, IN*

- Designed a multilingual NLU framework (intent + entity) integrated into DeHaat's chatbot ecosystem, improving query understanding accuracy by $\sim$ 25% compared to the previous rule-based version.
- Built domain-specific datasets and annotation schemas in Hindi, English, and regional languages, enabling scalable onboarding of new agricultural intents across product lines.
- Led architectural planning for future chatbot services, reducing model retraining time by 30–40% through modularized components and reusable embedding pipelines.
- Delivered documentation and benchmarking guidelines used by internal ML teams to standardize evaluation and rollout of new models.

Data Scientist — Jan. 2024 – Aug. 2024
DeHaat — *Gurgaon, IN*

- Built an LLM-powered agronomist assistant used internally to provide advisory to farmers, reducing manual search time for crop disease & soil-management information by >60%.
- Developed RLHF and DPO fine-tuning pipelines for open-source chat models, improving internal evaluation scores by $\sim$ 20–30% and reducing model hallucination in domain-specific queries.
- Created Flask APIs to productionize the assistant, enabling frictionless model deployment across frontend and backend teams with zero downtime rollouts.
- Automated data extraction using Selenium + BeautifulSoup, generating 10,000+ agricultural records from government advisories and agronomy portals, forming the foundation of a VectorDB retrieval layer.
- Integrated LangChain-based workflows and custom tools, speeding up experimentation cycles for prompt and agent iterations by $\sim$ 35%.

Junior Data Scientist — June 2023 – Jan. 2024
Nuvoretail — *Delhi, IN*

- Built and deployed transformer + random forest models to optimize search advertising, achieving 73% R^2 and enabling the marketing team to make 30–40% more accurate spend decisions.
- Developed a dynamic budget allocation algorithm that reduced overspending on low-performing keywords by $\sim$ 20%, improving ROI across Amazon ad campaigns.
- Engineered a scraping pipeline using Selenium to collect 50,000+ Amazon product listings, which became the primary dataset for market share analysis and ML experimentation.
- Produced actionable insights on SKU-level performance, empowering leadership to adjust pricing and bidding strategies based on market competitiveness.

Senior Project Assistant — April 2022 – Aug. 2022
Indian Institute of Technology — *Delhi, IN*

- Led an ICMR-funded research project to build a nerve monitoring prototype using custom microcontroller firmware, reducing latency of signal acquisition by $\sim$ 35%.
- Designed algorithms to assess skin impedance data, improving system stability and accuracy across varying contact conditions.
- Delivered technical reports, calibration results, and demo presentations, forming the basis for follow-up grant submissions and lab-scale deployments.

Internships

Machine Learning Intern Sep. 2021 – Nov. 2021
Knowledge Solutions India *Remote, IN*

- Developed SQL pipelines to extract and transform large relational datasets, reducing manual reporting time by >50%.
- Built regression and clustering models for client datasets, improving feature interpretability and decision-making for internal stakeholders.
- Conducted EDA and feature engineering that increased baseline model performance by $\sim$ 10–15%

Project Management Intern June 2019 – July 2019
Larsen & Toubro *Mumbai, IN*

- Updated and maintained Bill of Materials using Excel automation, improving engineering team visibility on inventory and reducing mismatch errors by $\sim$ 30%.
- Researched hydraulic systems and produced a comprehensive technical report, supporting training material for junior engineers.
- Collaborated with cross-functional teams to streamline project workflows, enhancing communication and efficiency.
- Assisted in scheduling and coordinating project meetings, ensuring timely follow-ups and documentation.

Education

The University of Manchester Manchester, UK
MSc in Data Science (Social Analytics) *Sep. 2024 – Sep. 2025*

- Modules: Statistics and Machine Learning, Topological Data Analysis, Cognitive Robotics and Computer Vision.
- **Bias Detection in LLMs:** Developed a multitask DeBERTa-based model to detect political, gender, and immigration bias in LLM outputs, introducing a novel metric for bias assessment. Evaluated and compared bias levels across multiple models to promote fairness and transparency in AI systems [Test Model].
- **Social Listening Platform (Vulse Collaboration):** Built a social listening and advisory web application that collects real-time data from Twitter (X), Facebook, and web sources via official APIs. Integrated LangChain-based LLM pipelines for competitor analysis and marketing recommendations. Developed with Next.js, Node.js, and MongoDB, featuring secure authentication, API documentation, and scalable architecture for enterprise deployment.

Imarticus Learning Delhi, IN
Post-Graduate Program in Data Analytics and Machine Learning *Aug 2022 – Apr 2023*

- Covered statistical foundations, Python and R for Data Science, SQL programming, and data visualization using Tableau and Power BI.
- **Neural Machine Translation using Transformer Architecture:**
  - Built a Transformer-based neural machine translation model for English to Hinglish, using TensorFlow and Keras with custom encoder-decoder architecture and positional embedding layers.
  - Preprocessed and vectorized text data using a custom CharBPE tokenizer via Hugging Face, creating efficient input-output pairs for training.
  - Trained and evaluated the model, optimized hyperparameters, and validated performance through real-world translation examples, documenting the full pipeline in a detailed presentation.
- **Stroke Prediction Model using PySpark on Hadoop Cluster in GCP:**
  - Developed and deployed a PySpark-based Random Forest model for stroke prediction on a GCP-hosted Hadoop cluster, achieving 81% accuracy.
  - Performed comprehensive EDA and preprocessing, including data cleansing, normalization, and feature engineering using PySpark ML libraries.
  - Managed GCP Hadoop cluster infrastructure, ensuring scalable computation, and delivered a detailed presentation highlighting methodologies, performance metrics, and insights.

University of Mumbai Mumbai, India
Bachelor of Engineering in Mechanical Engineering *July 2017 – Aug. 2021*

- Relevant Courses: Applied Mathematics, Applied Physics, Mechatronics, Operations Research, Industrial Electronics, Computer Programming in C and Database Management Systems.
- Dissertation — Quadruped Robot: *Designed and built a 12-DOF quadruped robot using SolidWorks and 3D printing; implemented inverse kinematics in C and Python for wireless control. Integrated reinforcement learning with ultrasonic sensors for autonomous obstacle avoidance. Project Links: PSybeR-Cat, Spider*

Publications

A Topology of Educational Outcome Inequality 2025
Benites, A., Hara, K., Singh, P., Zubier, B., & Rudkin, S. *SSRN ID: 5401815*

- Published on SSRN exploring topological approaches to understanding inequality in educational outcomes across social and economic dimensions.

Hackathons

AI Encode London Hackathon London, UK
1st Prize – StarkWare Bounty *Apr. 2025*

- Built a blockchain-based open-world game integrating Generative AI and crypto land trading. Recognized for innovation in combining scalable blockchain infrastructure with immersive AI-driven gameplay.

Projects

Tool Integration Engine (TIE) | *Python, LLMs, Knowledge Graphs, PyTorch Geometric* October 2025 – Present

- Designed and developed a modular engine that enables Large Language Models to dynamically discover and invoke external tools using graph-based reasoning. Tools, parameters, and domains are represented as nodes in a knowledge graph, allowing structured traversal for context-aware API selection. Supports plug-and-play registration for enterprise systems and integrates GNN-based models for learning optimal tool invocation paths.
- Implemented multi-relational graphs in NetworkX and trained GNNs using PyTorch Geometric for tool prediction. Future versions will incorporate reinforcement learning for dynamic routing and multi-step task orchestration.

MIRA (AI Operating System) | *Python, LLMs, RAG, ReactJS* September 2025

Personal Research Project

- Engineered a Linux-based AI Operating System capable of autonomously managing system operations through natural language commands.
- Enabled file system control, OS health monitoring, and task automation, allowing MIRA to manage files, check resource usage, and execute operations on demand.
- Integrated Google Calendar, Gmail, and Notes using OAuth 2.0 for context-aware scheduling, email handling, and note summarization.
- Built a real-time frontend with Vite + React, providing a conversational terminal UI to visualize system activity and AI responses.
- Developed a modular backend with FastAPI, LLM orchestration, and a secure Docker-based execution layer featuring guardrails and audit logging.
- Implemented retrieval-augmented generation (RAG) over Linux command documentation and personal data, enabling intelligent reasoning and contextual decision-making.
- Designed for scalability, privacy, and extensibility, supporting local or dedicated server deployment for a fully autonomous AI OS experience.

ZeroMQ Communication Patterns (Educational Project) | *Python, ZeroMQ* Jun. 2023

- Developed an educational project to teach the five core ZeroMQ messaging patterns (REQ/REP, PUB/SUB, PUSH/PULL, ROUTER/DEALER, PAIR) using Python (`pyzmq`).
- Authored step-by-step, runnable examples demonstrating asynchronous and distributed communication across microservices and network nodes.
- Focused on simplifying complex concepts like socket management, concurrency, and non-blocking I/O for learners and developers.
- Designed reusable templates and clear documentation to serve as a learning resource for distributed systems and IPC fundamentals.
- Published publicly to help others understand ZeroMQ's core communication models. GitHub Link

LazyForecast: A Python Library | *Python, GitHub Actions, TensorFlow, Machine Learning* Apr. 2023

- Developed and published **LazyForecast** on PyPI — a Python library for lazy time-series forecasting, supporting models such as ARIMA, MLP, and LSTM. It automates data preprocessing, model training, and evaluation with minimal code.
- Integrated auto-ARIMA via `pmdarima` for automatic hyperparameter optimization, and added visualization utilities for forecast diagnostics and interpretability.
- Set up GitHub Actions for CI/CD, automating PyPI releases and ensuring robust package deployment. PyPI Link, GitHub Repo

Achievements

- Delivered Guest Lecture at the Ministry of Defence (2023): Presented on AI concepts and conducted a hands-on training session for MoD employees.
- Cleared GATE (2024): Qualified the Graduate Aptitude Test in Engineering for the Data Science and Artificial Intelligence specialization.
- Scored 7.5 Bands in IELTS (Mar 2024): Demonstrated strong English proficiency across listening, reading, writing, and speaking.
- Led Industrial Visit to Indian Ordnance Factory (2019): Organized and coordinated a successful visit for the Mechanical Engineering Department.
- Bronze Medal – National Science Olympiad (NSO) (Dec 2014): Recognized for academic excellence at the national level in science.

Certificates

- Azure AI Fundamentals – AI900 by Microsoft
- Machine Learning with AI using Python by Certiport
- Microsoft AI Classroom Series by Microsoft
- Introduction to Big Data by UC San Diego via Coursera
- Machine Learning: Regression by University of Washington via Coursera
- Python Data Structures by University of Michigan via Coursera
- AI for Everyone by Deeplearning.ai via Coursera
- A1 – German (Deutsch) Language via Goethe Institut Indien

Workshops

- Attended a 2-day workshop on "Machine Learning" at IIT Bombay (24–25 Aug 2019).

Technical Skills

Languages: Python, C/C++, SQL, JavaScript, TypeScript, HTML/CSS, R
Frameworks: React, Node.js, Flask, Material-UI, FastAPI, DJango
Databases: MS SQL, MySQL, Hive, GraphDB
Developer Tools: Git, Docker, Google Cloud Platform (GCP), Amazon Web Services (AWS), VS Code, Visual Studio, GitHub Actions, Linux, Sagemaker, Bedrock, Hadoop
Libraries: Pandas, NumPy, Matplotlib, Seaborn, Sci-kit Learn, TensorFlow, Keras, OpenCV, PySpark, BeautifulSoup, Selenium, YOLO, PyTorch, LangChain, LangGraph
Other: Advance Excel, Tableau, LLM, RAG, , Agentic-AI, Reinforcement Learning, GraphDB, Graph Neural Network (GNN), ZeroMQ (pyzmq)

Soft Skills

Business Acumen **Problem-Solving** **Collaboration** **Adaptability** **Communication**

Expertise

Machine Learning
Deep Learning
Statistical Analysis
Data Visualization
Data Preparation
Data Structures & Algorithms
Data Exploration
Natural Language Processing
Computer Vision